CARSON SLATER

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EDUCATION

PhD Statistical Science, Baylor University 2024 - 2027 [Expected]
Advised by Amanda S. Hering · Concentrations in *Data Science* and *Biostatistics*
Research involves developing unsupervised learning methods for time series with applications in municipal water distribution.

M.S. Statistical Science, Baylor University 2023 - 2024
(GPA: 3.86/4.0)

B.S. Applied Mathematics with Statistics, Wheaton College (IL) 2019 - 2023
B.A. Economics
(GPA: 3.46/4.0), *Edman Presidential Scholarship*
Varsity Baseball Athlete (NCAA-DIII)

EXPERIENCE

Data Analyst August 2026 - Present
Texas Farm Bureau Insurance Companies, *Part Time* *Waco, TX*

- Supporting actuarial teams in developing and deploying property and casualty models for risk assessment and pricing.

Research Assistant January 2024 - July 2026
Baylor University *Waco, TX*

- Developed statistical methods for assessing urban water consumption with IoT meters. Research is funded by Denver Water.
- Extracted, transformed, and loaded unstructured data for analyses for Denver Water collaborators.
- Collaborated with domain experts in urban water demand planning to assess water conservation interventions, implementing rigorous hypothesis testing and building Shiny dashboards to visualize individual and aggregate customer trends.
- Engineered features from household-level billing and water usage time series to perform market segmentation for Denver Water's single-family residential customer base, coming up with seven user personas to characterize their population (∼200k homes).
- Edited and reviewed material for an undergraduate intro to data science course. (DOI: 10.7910/DVN/BZJW2E)

Statistical Consultant August 2023 - December 2025
Baylor University *Waco, TX*

- Created and assessed a quantitative framework for assessing the success of college athletics coaches for the Baylor Athletic director. Web scraped and compiled a large dataset for modeling collegiate coach success.
- Consulted for ongoing interdisciplinary research in the computer engineering and environmental science departments, and for the American Holistic Nursing Association. Published academic projects in peer-reviewed journals.
- Delivered quality reports for each client and had regular meetings to discuss the project, the methodology and results.

Academic Coach January - May 2023
Wheaton Academy Center for Lifelong Learning *West Chicago, IL*

- Mentored students 1-on-1 to sharpen math skills from Algebra I to AP Statistics.

Mathematics Teaching Assistant August 2022 - May 2023
Wheaton College *Wheaton, IL*

- Facilitated help sessions for undergraduate students in the topics of Machine Learning and Probability Theory.

SKILLS

Technology	Docker, Positron (≈ VS Code), RStudio, Git/Github, reveal.js, SQL, duckDB, LaTeX
Python	pandas, numpy, PyTorch (Lightning), plotnine, seaborn, plotly, scikit-learn, Jupyter
R	tidyverse (dplyr, ggplot2, etc.), TensorFlow (Keras), torch, shiny, RMarkdown/Quarto, rstan, rjags, caret, tidymodels, targets (gittargets, proffer, etc.)
Stats & ML	statistical modeling, hypothesis testing, mixture models, time series, regression, linear mixed models, clustering, classification, deep learning, PCA, random forests, ensemble models (e.g., XGBoost), SVM, Bayesian methods, spatiotemporal modeling, convolutional neural networks, denoising diffusion probabilistic models

LEADERSHIP EXPERIENCE

President May 2026 - Present
Baylor University Statistical Science Paper Club *Waco, TX*

- Sourcing and facilitating discussions on frontier papers in statistics, biostatistics, and machine learning for graduate students in the statistical science department.

President March 2024 - May 2025
Baylor University Sports Analytics Club *Waco, TX*

- Led a team of undergraduates in submitting a March Madness bracket challenge submission on Kaggle.
- Taught undergraduates basic version control, R programming, statistical and machine learning topics.